

# Modularity, cyclicity, and constrained nonuniformity in Spanish stress assignment\*

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## Abstract

We develop a stratal analysis of Spanish stress in which morphosyntactic conditioning and metrical nonuniformity are tightly constrained. Morphosyntactic conditioning is limited to cyclicity, the distinction between stems and affixes, and the affiliation of affixes to the stem or word level. Nonuniformity arises from lexical prespecification and interstratal constraint reranking. This framework yields several successful empirical predictions: Richness of the Base requires that prespecification be unpredictably distributed across all morph types; Chung’s Generalization demands identical metrical patterns in mono- and polycyclic stem-level constructions; and the Monomorpheme Generalization entails that monomorphemic and overtly affixed stem-level items also exhibit the same patterns.

**Keywords:** modularity, cyclicity, nonuniformity, prespecification, stress, Spanish.

## 1 Introduction

This was all; and this was enough.

Macaulay

Spanish stress assignment involves a complex interaction of phonological, morphological, and lexical factors. In (1a), for example, the distinction between oxytonic stress in *caríz* and paroxytonic stress in *cára* instantiates the default phonological pattern: stress is attracted to a domain-final heavy syllable. In (1b), in contrast, the heavy ultima of *cára-s* remains unstressed owing to a cyclic

effect: the plural form preserves the stress profile of the uninflected stem contained within it. In turn, the paroxytonic contour of *lápiz* (1c), where the final heavy again fails to receive stress, reflects lexical prespecification.

- |     |    |               |           |           |
|-----|----|---------------|-----------|-----------|
| (1) | a. | <i>cára</i>   | [ˈka.ra]  | ‘face’    |
|     |    | <i>caríz</i>  | [ka.ˈriθ] | ‘aspect’  |
|     | b. | <i>caríz</i>  | [ka.ˈriθ] | ‘aspect’  |
|     |    | <i>cára-s</i> | [ˈka.ras] | ‘face-PL’ |
|     | c. | <i>caríz</i>  | [ka.ˈriθ] | ‘aspect’  |
|     |    | <i>lápiz</i>  | [ˈla.piθ] | ‘pencil’  |

The interaction between the default phonological pattern, cyclic effects, and lexical prespecification gives rise to intricate patterns, addressed in an extensive literature (e.g., Hooper and Terrell 1976; Harris 1983, 1995; Roca 1988, 2005, 2020; Halle, Harris and Vergnaud 1991; Oltra-Massuet and Arregi 2005; Baković 2016). Faced with this challenge, many previous analyses have resorted to highly permissive conceptions of the morphosyntax-phonology interface. For example, Oltra-Massuet and Arregi's (2005) influential treatment of verb stress abandons modularity by allowing phonological rules to align metrical constituents with specific functional heads in the syntax. Relatedly, Baković (2016) posits a domain of stress assignment that consists of the stem minus its theme vowel, though it can be shown that this does not coincide with any independently motivated morphosyntactic constituent (see section 5).

In this article, in contrast, we show that the facts of Spanish stress submit to a stratal analysis (Kiparsky 2000, 2015; Bermúdez-Otero 2011, 2012, 2018b; Trommer 2025) in which both phonological access to morphosyntactic information and nonuniform behavior within the phonology itself are tightly controlled by general theoretical principles. More specifically, morphosyntactic

conditioning is limited to three factors. First, the phonology applies cyclically over a hierarchy of domains projected from morphosyntactic structure in accordance with a general procedure. Secondly, this procedure refers to the idiosyncratic specification of individual affixes as belonging to the stem or word level. Thirdly, constraints on prosodic structure are sensitive to the broad distinction between lexical and functional constituents, but do not refer to specific lexical categories, syntactic features, morphemes, or lexical items. Thus the analysis maintains a high standard of modularity (Bermúdez-Otero 2012, Trommer 2015; cf. Scheer 2012).

In turn, phonological nonuniformity is traced to just two sources: lexical prespecification and interstratal differences. Both are tightly constrained. By Richness of the Base (Prince and Smolensky [1993] 2004:205, 225; McCarthy 2002:69-82), lexical prespecification is inert unless protected by high-ranking faithfulness constraints in the stem-level phonology, and active prespecifications are required to be unpredictably distributed across all morphs, be they stems, derivational affixes, or inflectional affixes of whatever category. Interstratal differences are also limited insofar as they reduce to a binary choice between the stem-level and the word-level phonology; within each stratum, a single constraint ranking holds. In consequence, the same set of metrical possibilities is available to all stem-level constructions, whether monomorphemic (2a) or suffixed (2b,c), and monocyclic (2a,b) or polycyclic (2c).

| (2)                |                                                                        |                                                    | Suffixed? | Polycyclic? |
|--------------------|------------------------------------------------------------------------|----------------------------------------------------|-----------|-------------|
| a. stem            | $[\text{SL } \textit{azúcar} ]$                                        | $[\text{a.}'\theta\text{u.kar}]$<br>'sugar'        | no        | no          |
| b. root + SL affix | $[\text{SL } \textit{vers-átıl} ]$                                     | $[\text{ber.}'\text{sa.tıl}]$<br>'versatile'       | yes       | no          |
| c. stem + SL affix | $[\text{SL } [\text{SL } \textit{tolera} ] [\text{SL } \textit{s} ] ]$ | $[\text{to.}'\text{le.ras}]$<br>'tolerate.PRS.2SG' | yes       | yes         |

Beyond its modularity and its strict control of nonuniformity, the analysis enjoys two additional conceptual advantages. First, it assumes a very simple and general procedure for defining cyclic domains: every minimal constituent specified for a syntactic category (minimal stem or affix) goes through a cycle of the stem-level phonology; combinations of a stem plus a stem-level affix also trigger stem-level cycles; and, finally, fully inflected grammatical words define cycles for the word-level phonology. This contradicts claims by Cole (1995) and by Myler (2015), who wholly or partially deny stem-level cyclic recursion in Spanish. Secondly, metrical prespecification relies exclusively on genuine elements of phonological representation that are independently motivated; *ad hoc* extrametricality diacritics (Den Os and Kager 1986) or parochial nonfinality constraints (Baković 2016) are avoided.

This tightly restricted theoretical apparatus generates a rich set of empirical predictions. We will show that our analysis demarcates default from exceptional stress patterns correctly, as demonstrated by lexical statistics (Núñez-Cedeño and Morales-Front 1999), neologisms and loanword adaptation (Bermúdez-Otero 2007:258-262; Torres-Tamarit and Martínez-Paricio 2024), laboratory experiments (Waltermire 2004, Fuchs 2018), and evidence from historical change (Martínez-Paricio 2013). In addition, we are the first to formulate and confirm the three predictions stated above: that Spanish exhibits the same set of underlying metrical prespecifications in morphs of all types (Richness of the Base), and that the same inventory of metrical patterns occurs in polycyclic as in monocyclic stem-level constructions (Chung's Generalization), and in overtly affixed as in monomorphemic stem-level constructions (Monomorpheme Generalization).

These empirical advantages contrast with the descriptive shortcomings incurred by less constrained analyses. For example, Baković's (2016) appeal to stems shorn of their theme vowels overgeneralizes the exceptional oxytonic contour of nouns such as *rubí* [ru.ˈβi] ‘ruby’, thereby mispredicting the behavior of acronyms like *ÚCI* [ˈu.θi] ‘intensive care unit’, cf. \**UCÍ*. Similarly,

Oltra-Massuet and Arregi (2005) fail to predict the direction of ongoing analogical change in the stress pattern of present-tense verbs: *cantémos* [kan. 'te.mos] 'sing.PRS.SBJV.1PL' > *cántemos* ['kan.te.mos] (e.g., in Uruguayan Spanish: see Martínez-Paricio 2013), but *cánten* ['kan.ten] 'sing.PRS.SBJV.3PL' ≠ *\*cantén* [kan. 'ten].

The exposition proceeds as follows. In section 2 we show that the default metrical pattern of Spanish consists of a bimoraic trochee aligned with the right edge of the phonological domain; within the final trisyllabic window, however, this pattern may be overridden by underlying prespecifications protected by high-ranking faithfulness constraints in the stem-level hierarchy. Section 3 addresses the metrical behavior of morphologically complex items. We first lay out the general theory of cyclic domain assignment, adducing evidence from high vocoid syllabification to show that stem-level cyclic domains are recursive in Spanish as in other languages (section 3.1). One of our principles of cyclic domain assignment asserts that each affix goes through an individual cycle of the stem-level phonology; we show that, in those cycles, affixes exhibit default behavior if allowed by their size, unless they bear one of the permitted underlying prespecifications (section 3.2). Combinations of a stem with one or more stem-level affixes are then shown to bear stress on the rightmost accented syllable in the input to the relevant cycle; this simple mechanism accounts in full for the metrical patterns observed in inflected verbs, including the neutralization of root stress contrasts and the prevalence of columnar stress (section 3.3). Combinations of a stem with one or more word-level affixes behave in exactly the same way, except that the final three-syllable window is turned off at the word level (section 3.4). Section 4 recapitulates the theoretical foundations of the analysis and corroborates the three empirical predictions stated above: namely, those following from Richness of the Base, Chung's Generalization, and the Monomorpheme Generalization. Section 5 goes on to diagnose the incorrect generalizations in Baković (2016) and Oltra-Massuet and Arregi (2005), which are shown to be made available by an excessively permissive conception of the morphosyntax-phonology interface. Our results are summarized in section 6.

## 2 Default footing and lexical prespecification

### 2.1 The default pattern

As already illustrated in (1a), default stress in Spanish is oxytonic if the ultima is heavy, else paroxytonic. This pattern emerges very robustly in wordform type frequencies, even without controlling for cyclic effects in morphologically complex forms (Núñez-Cedeño and Morales-Front 1999:211, discussed in Bermúdez-Otero 2007:258). It also holds over several types of items that exhibit no evidence of internal morphological structure. First, it clearly predominates in the citation forms of function words: these normally bear stress on the ultima when consonant-final (3a) and on the penult when vowel-final, even though final vowels in these forms are not plausibly analysed as thematic, either because of their segmental quality (3b) or, more generally, because they are not involved in the exponence of inflectional features (3c) (see further section 5).

- |     |    |                |             |                |
|-----|----|----------------|-------------|----------------|
| (3) | a. | <i>según</i>   | [se.ˈɣun]   | ‘according to’ |
|     |    | <i>también</i> | [tam.ˈbjen] | ‘also’         |
|     |    | <i>jamás</i>   | [xa.ˈmas]   | ‘never’        |
|     | b. | <i>cási</i>    | [ˈka.si]    | ‘almost’       |
|     | c. | <i>pára</i>    | [ˈpa.ra]    | ‘for’          |
|     |    | <i>cuándo</i>  | [ˈkwan.do]  | ‘when’         |
|     |    | <i>sóbre</i>   | [ˈso.βre]   | ‘upon’         |

Secondly, it systematically applies to acronyms (Torres-Tamarit and Martínez-Paricio 2024):

- (4) a. *ÚCI* [ˈu.θi] ‘intensive care unit’  
*ÓNU* [ˈo.nu] ‘UN’  
 b. *UNICÉF* [u.ni.ˈθef] ‘UNICEF’  
*Repsól* [rep.ˈsol] corporation name

Thirdly, the default pattern also generally extends to loanwords (5a), except in the case of hyperforeignisms (5b). In the latter, deviant stress functions as a marker of foreignness. This is confirmed by two observations. First, the status of such loanwords as alien is reflected in their inflectional behavior: they are systematically excluded from native nominal classes, and so behave as athematic (Bermúdez-Otero 2007:261). Secondly, their stress contours need not match the source language; a similar phenomenon is observed in English pronunciations such as *Medici* for Italian *Médici* (Dabouis and Fournier 2022).

- (5) a. *futón* [fu.ˈton] ‘futon’ (note *e*-stem plural *futón-e-s*)  
*wasábi* [gwa.ˈsa.βi] ‘wasabi’  
 b. *Ámsterdam* [áms.ter.ðam] ‘Amsterdam’ cf. Dutch [ˌamstərˈdam]  
*Hálloween* [xá.lo.ɣwin] ‘Halloween’ cf. English [ˌhæləʊˈiːn]

Finally, the default status of oxytonic stress over heavy ultimas, and of paroxytonic stress elsewhere, has been experimentally corroborated in controlled wug tests (Waltermire 2004:185).

This default pattern submits to a straightforward analysis in which a bimoraic trochee is aligned with the right edge of the visible domain. We implement this analysis by means of the constraints in (6a-d). In tableau (6e) we use the *o*-class noun *camín-o* [ka.ˈmi.no] ‘path’ to illustrate the assignment of default paroxytonic stress to a vowel-final stem, and the athematic noun *azimút* [a.θi.ˈmut] ‘azimuth’ to illustrate the assignment of default oxytonic stress to a consonant-final

stem. The plural form *azimút-s* [a.θi.'muts] ‘azimuths’ confirms that the stem *azimút* is athematic; this explains why a theme vowel is absent even though /t/ is phonotactically dispreferred in domain-final position. In fact, most Spanish nouns with singular forms ending in a consonant behave differently: they belong to the ordinary *e*-stem class, as indicated by their plural forms (e.g. *caracól-Ø* [ka.ra.'kol] ‘snail.SG’, *caracól-e-s* [ka.ra.'ko.les] ‘snail-PL’). The thematic element of this class consists of the underlying disjunction /-{Ø,e}/, which is resolved in the phonology (Bermúdez-Otero 2013:9-17). The null allomorph is selected when the thematic element is immediately preceded by one of the following singleton consonants, /ɾ, l, n, s, θ, d/, which are phonotactically unmarked in domain-final position. In tableau (6f), this is illustrated with the example of *caracól-Ø*; the preference for the null allomorph of the theme is driven by a constraint against domain-final unstressed [e], here labeled \*ě]. In contrast, when the theme is preceded by a consonant that would be phonotactically marked in domain-final position, the overt exponent /-e/ is preferred. In tableau (6f), this is illustrated with the example of *tomát-e* [to.'ma.te] ‘tomato’; in this case, the constraint against domain-final [t], here labeled \*t], compels a violation of lower-ranked \*ě]. Theme allomorph selection and foot construction interact transparently, and so tableau (6f) shows them as taking place in parallel within the same stem-level cycle.

- (6)
- a. FTBIN(μ): A foot must be exactly bimoraic.
  - b. ALIGN-R(Ft,D): The right edge of a foot must coincide with the right edge of the domain.
  - c. TROCH: A foot-head must be initial in its foot (Martínez-Paricio and Kager 2015:473).
  - d. IAMB: A foot-head must not be initial in its foot (Martínez-Paricio and Kager 2015:473).



e. *camín-o* [ka.'mi.no] ‘path’, *azimút* [a.θi.'mut] ‘azimuth’

|         |                                                      | TROCH | FTBIN(μ) | ALIGN-R(Ft,D) | IAMB |
|---------|------------------------------------------------------|-------|----------|---------------|------|
| kamin-o | → ka('mi <sup>μ</sup> .no <sup>μ</sup> )             |       |          |               | *    |
|         | ka(mi <sup>μ</sup> .'no <sup>μ</sup> )               | *W    |          |               | L    |
|         | ('ka <sup>μ</sup> .mi <sup>μ</sup> )no               |       |          | *W            | *    |
|         | ka.mi('no <sup>μ</sup> )                             |       | *W       |               | *    |
| aθimut  | → a.θi('mu <sup>μ</sup> t <sup>μ</sup> )             |       |          |               | *    |
|         | a(θi <sup>μ</sup> .'mu <sup>μ</sup> t <sup>μ</sup> ) | *W    | *W       |               | L    |
|         | ('a <sup>μ</sup> .θi <sup>μ</sup> )mut               |       |          | *W            | *    |
|         | a('θi <sup>μ</sup> .mu <sup>μ</sup> t <sup>μ</sup> ) |       | *W       |               | *    |
|         | a('θi <sup>μ</sup> )mut                              |       | *W       | *W            | *    |

f. *caracól-Ø* [ka.ra.'kol] ‘snail’, *tomát-e* [to.'ma.te] ‘tomato’

|               |                                                       | *t] | *ě] | TROCH | FTBIN(μ) | ALIGN-R(Ft,D) | IAMB |
|---------------|-------------------------------------------------------|-----|-----|-------|----------|---------------|------|
| karakol-{Ø,e} | → ka.ra('ko <sup>μ</sup> l <sup>μ</sup> )             |     |     |       |          |               | *    |
|               | ka(ra <sup>μ</sup> .'ko <sup>μ</sup> l <sup>μ</sup> ) |     |     | *W    | *W       |               | L    |
|               | ('ka <sup>μ</sup> .ra <sup>μ</sup> )kol               |     |     |       |          | *W            | *    |
|               | ka('ra <sup>μ</sup> .ko <sup>μ</sup> l <sup>μ</sup> ) |     |     |       | *W       |               | *    |
|               | ka('ra <sup>μ</sup> )kol                              |     |     |       | *W       | *W            | *    |
|               | ka.ra('ko <sup>μ</sup> .le <sup>μ</sup> )             |     | *W  |       |          |               | *    |
|               | ka.ra(ko <sup>μ</sup> .'le <sup>μ</sup> )             |     | *W  | *W    |          |               | L    |
|               | ka('ra <sup>μ</sup> .ko <sup>μ</sup> )le              |     | *W  |       |          | *W            | *    |
|               | ka.ra.ko('le <sup>μ</sup> )                           |     | *W  |       | *W       |               | *    |
| tomat-{Ø,e}   | to('ma <sup>μ</sup> t <sup>μ</sup> )                  | *W  | L   |       |          |               | *    |
|               | (to <sup>μ</sup> .'ma <sup>μ</sup> t <sup>μ</sup> )   | *W  | L   | *W    | *W       |               | L    |
|               | ('to <sup>μ</sup> .ma <sup>μ</sup> t <sup>μ</sup> )   | *W  | L   |       | *W       |               | *    |
|               | ('to <sup>μ</sup> )mat                                | *W  | L   |       | *W       | *W            | *    |
|               | → to('ma <sup>μ</sup> .te <sup>μ</sup> )              |     | *   |       |          |               | *    |
|               | to(ma <sup>μ</sup> .'te <sup>μ</sup> )                |     | *   | *W    |          |               | L    |
|               | ('to <sup>μ</sup> .ma <sup>μ</sup> )te                |     | *   |       |          | *W            | *    |
|               | to.ma('te <sup>μ</sup> )                              |     | *   |       | *W       |               | *    |

## 2.2 Lexically prespecified patterns

We now turn to nondefault stress patterns. We analyze these as being caused by underlying metrical prespecification. Crucially, as anticipated in section 1, prespecification is under grammatical control: by Richness of the Base, inputs may bear any metrical specification compatible with the general theory of phonological representations, but only those specifications that are protected by high-ranking faithfulness constraints survive. We will establish the inventory of legal nondefault patterns with examples from monomorphemic items (*modulo* theme vowels), where cyclic effects are known to be absent.

First, oxytonic stress is marked when the ultima is light. This nondefault specification is protected by the faithfulness constraint IDENT(stress) (7a). We assume that the result in such cases is a degenerate foot (7b). We know that such feet are allowed because of the existence of stressed lexical monosyllables such as *fě* ['fe] 'faith' (8b). The fact that such monosyllables are licit indicates that FTBIN( $\mu$ ) ranks below the requirement that lexical items should project a prosodic word (8a); this prosodic word must in turn contain a foot in order to satisfy the inviolable principle of Proper Headedness (Ito and Mester 2003:37).

- (7) a. IDENT(stress): An output segment matches its input correspondent for stress or lack of stress.

b. *canapé* [ka.na.'pe] 'canapé'

| kana'pe                    | IDENT(stress) | TROCH | FTBIN( $\mu$ ) | IAMB |
|----------------------------|---------------|-------|----------------|------|
| → ka.na('pe $^\mu$ )       |               |       | *              | *    |
| ka(na $^\mu$ .'pe $^\mu$ ) |               | *W    | L              | L    |
| ka('na $^\mu$ .pe $^\mu$ ) | *W            |       | L              | *    |

- (8) a. LEX↔PWD: Every maximal node containing nonfunctional (root) material must correspond to a prosodic word, and vice versa (Prince and Smolensky [1993] 2004).

b. *fě* ['fe] 'faith'

| fe                               | LEX↔PWD | FtBIN(μ) |
|----------------------------------|---------|----------|
| → (ω (Ft (σ 'fe <sup>μ</sup> ))) |         | *        |
| (σ fe <sup>μ</sup> )             | *W      | L        |

In turn, paroxytonic stress is marked when the ultima is heavy: for example, *azúcar*-Ø [a. 'θu.kar] 'sugar'. As shown in (9), this gives rise to a dispreferred trochee with a heavy dependent syllable, [a('θu.kar)], in violation of the constraint \*('LH). *Azúcar*-Ø is an ordinary *e*-stem, and so the tableau shows the selection of the null allomorph of its thematic element taking place in parallel with foot construction, just as in (6f).

(9) a. \*('LH): A trochee must not consist of a sequence of a light plus a heavy syllable (Prince 1990, Hayes 1995).

b. *azúcar*-Ø [a. 'θu.kar] 'sugar'

| aθúkar- {Ø,e}                                          | *ě] | IDENT(stress) | FtBIN(μ) | ALIGN-R(Ft,D) | *('LH) |
|--------------------------------------------------------|-----|---------------|----------|---------------|--------|
| → a('θu <sup>μ</sup> .ka <sup>μ</sup> r <sup>μ</sup> ) |     |               | *        |               | *      |
| a('θu <sup>μ</sup> )kar                                |     |               | *        | *W            | L      |
| a.θu('ka <sup>μ</sup> r <sup>μ</sup> )                 |     | *W            | L        |               | L      |
| a('θu <sup>μ</sup> .ka <sup>μ</sup> )re                | *W  |               | L        | *W            | L      |
| a.θu('ka <sup>μ</sup> .re <sup>μ</sup> )               | *W  | *W            | L        |               | L      |

Finally, all proparoxytones are marked because they are forced to leave the ultima unfooted, in violation of ALIGN-R(Ft,D); this is illustrated in tableau (10b). Moreover, the presence and distribution of heavy syllables induces markedness asymmetries among different types of proparoxytone. Least deviant are cases such as *sábana* ['sa.ba.na] 'bedsheet', comprising only light syllables. Items such as *régimen* ['re.xi.men] 'diet, regime' and *hipérbaton* [i.'per.βa.ton] 'hyperbaton' are far rarer, as they contain a heavy ultima in violation of WSP (10a). The most exceptional cases are those with a heavy penult, like *Frómista* ['fro.mis.ta] (place name): these

violate both WSP and \*('LH). The resulting relationships of harmonic bounding give rise to the descending harmony scale *sábana* > *régimen* > *Frómista* (10c), which is reflected in lexical type frequencies and the results of nonce-word rating tasks (Fuchs 2018).

(10) a. WSP: Heavy syllables are stressed.

b. *Frómist-a* ['fro.mis.ta] (place name)

| frómist-a                                                | IDENT(stress) | FtBIN( $\mu$ ) | ALIGN-R(Ft,D) | *('LH) | WSP |
|----------------------------------------------------------|---------------|----------------|---------------|--------|-----|
| → ('fro <sup>μ</sup> .mi <sup>μ</sup> s <sup>μ</sup> )ta |               | *              | *             | *      | *   |
| fro('mi <sup>μ</sup> s <sup>μ</sup> .ta <sup>μ</sup> )   | *W            | *              | L             | L      | L   |
| fro('mi <sup>μ</sup> s <sup>μ</sup> )ta                  | *W            | L              | *             | L      | L   |

c. Harmonic bounding in proparoxytones

|               | ('sa <sup>μ</sup> .ba <sup>μ</sup> )na <sup>μ</sup> | ('re <sup>μ</sup> .xi <sup>μ</sup> )me <sup>μ</sup> n <sup>μ</sup> | ('fro <sup>μ</sup> .mi <sup>μ</sup> s <sup>μ</sup> )ta <sup>μ</sup> |
|---------------|-----------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| *('LH)        |                                                     |                                                                    | *                                                                   |
| WSP           |                                                     | *                                                                  | *                                                                   |
| ALIGN-R(Ft,D) | *                                                   | *                                                                  | *                                                                   |

As we shall see in section 3.4, violations of the final trisyllabic window occur only in certain morphologically complex word-level items. Absent such stratal effects, the final three-syllable window tolerates no exceptions. We model this restriction by subordinating IDENT(stress) to the markedness constraint \*LONGLAPSE (11a) in the stem-level phonology: this has the effect of forbidding domain-final sequences of more than one unfooted syllable.

(11) a. \*LONGLAPSE: An unstressed syllable must be adjacent either to a stressed syllable or to a domain edge (Kager 2007:213).

b. hypothetical /máripoſ-a/

| máripoſ-a                                   | *LONG LAPSE | IDENT(stress) | ALIGN-R(Ft,D) |
|---------------------------------------------|-------------|---------------|---------------|
| → ma.ri('po <sup>u</sup> .sa <sup>u</sup> ) |             | *             |               |
| ma.('ri <sup>u</sup> .po <sup>u</sup> )sa   |             | *             | *W            |
| ('ma <sup>u</sup> .ri <sup>u</sup> )po.sa   | *W          | L             | *W            |

### 3 Stress in morphologically complex items

#### 3.1 Cyclicity and stratification

We now turn to stress assignment in morphologically complex items. We assume that this proceeds cyclically over a hierarchy of domains projected from morphosyntactic structure in accordance with the general principles stated in (12).

(12) a. Every minimal constituent specified for a syntactic category (stem or affix) defines a cyclic domain for the stem-level phonology.

b. Every combination of a stem with a stem-level affix defines a cyclic domain for the stem-level phonology.

c. Every grammatical word (fully inflected lexical item) defines a cyclic domain for the word-level phonology.

In this view, acategorial roots do not trigger cycles (Kiparsky 1982a:144-145, 1982b:32-33; Inkelas 1989: section 3.5.5). Principle (12a) thus takes scope over two sets of items: the first comprises minimal stems, that is, stems immediately derived from roots; the second consists of affixes. The property which these two types of item share, and which opposes them to acategorial roots, is the fact that they are minimal and yet specified for some projecting syntactic category, whether lexical (*n*, *v*, *a*) or inflectional (T, ASP, NUM, etc.). Note, however, that Spanish theme vowels resemble roots rather than affixes in lacking a syntactic category: like a root, therefore, a theme vowel is incorporated into the cyclic domain defined by the affix to which it adjoins.



Phonology and Morphology (Mohan 1986:37-39). This insight has been partly reinstated in work within constraint-based Stratal Phonology that posits stem-level cycles over word-level affixes (Baker 2005, Buckler 2009, Bermúdez-Otero 2018b:111-112). This research notes that affixes sometimes behave like small stems: for example, German diminutive *-chen* [çən] exhibits the domain-initial palatal allophone of the dorsal fricative /x/, just like the stem *China* ['çi:na] ‘China’; similarly, in languages like Diyari (Poser 1989:127-128) and Ngalakgan (Baker 2005:4ff) every productive suffix constitutes a separate stress-assignment domain. Moreover, the stem-level phonology regulates the basic inventory of primary contrasts in a language, and so the observation that minimal stems and affixes never exhibit disjoint phonemic inventories is explained by the fact that both go through stem-level cycles with the same constraint hierarchy (cf. McCarthy 2007:133-134). Accordingly, we here restore Mohan’s insight in full by positing stem-level cycles over all affixes. In this respect our proposal agrees with Trommer’s (2011:73-77) Egalitarian Stratal OT, but without postulating a root stratum.

Needless to say, principle (12a) is fully consistent with extensive evidence showing that stems and affixes differ prosodically. As we shall see in section 3.2 below, this prosodic divergence follows from the fact that  $LEX \leftrightarrow PWD$  (8a) prevents affixes from projecting prosodic words. Principle (12a) is also fully consistent with the assertion that there exists phonologically driven allomorph selection by output optimization operating upon disjunctive input representations (Kager 1996, Mascaró 1996). In Spanish, for example, an idiosyncratic subset of third-conjugation verbs alternate between high and mid root-vowels: Bermúdez-Otero (2016) argues that, in this case, the morphology submits two listed stem allomorphs to the phonology, which prefers outputs with unmarked high vowels unless these incur OCP violations. Crucially, Bermúdez-Otero’s analysis has each stem pass separately through its own cycle of the stem-level phonology before competing with the other stem in the next phonological cycle: see the domain structure in (14). (The specification of verbal theme vowels as preaccenting is discussed in section 3.2 below.)

(14) *pido* ['pi.ðo] ‘ask.PRS.1SG’ (Bermúdez-Otero 2016:402-403)

$$\llbracket_{\text{SL}} \left\{ \begin{array}{l} \llbracket_{\text{SL}} \text{pid} - \acute{\sigma} \text{i} \rrbracket \\ \llbracket_{\text{SL}} \text{ped} - \acute{\sigma} \text{i} \rrbracket \end{array} \right\} \llbracket_{\text{SL}} \text{o} \rrbracket \rrbracket$$

The same approach straightforwardly generalizes to competition between affix allomorphs.

In turn, principle (12b) captures the observation that only stem-level domains, and not word- or phrase-level domains, are recursive. Once more, this insight originates in the early days of rule-based Lexical Phonology and Morphology, where the highest stratum was designated as *cyclic*, and later strata were designated as respectively *postcyclic* and *postlexical* (Booij and Rubach 1987; Halle and Vergnaud 1987). Why the three strata should differ from one another in this way is a lively topic of research: for example, see Bermúdez-Otero 2012:19-20, 31-40 on stem-level recursion, and compare Ishihara 2007 and Downing 2010 on putatively cyclic prosodification at the phrase level. We need not enter into this debate here, but we shall merely highlight the empirical evidence which shows that principle (12b) does hold for Spanish.

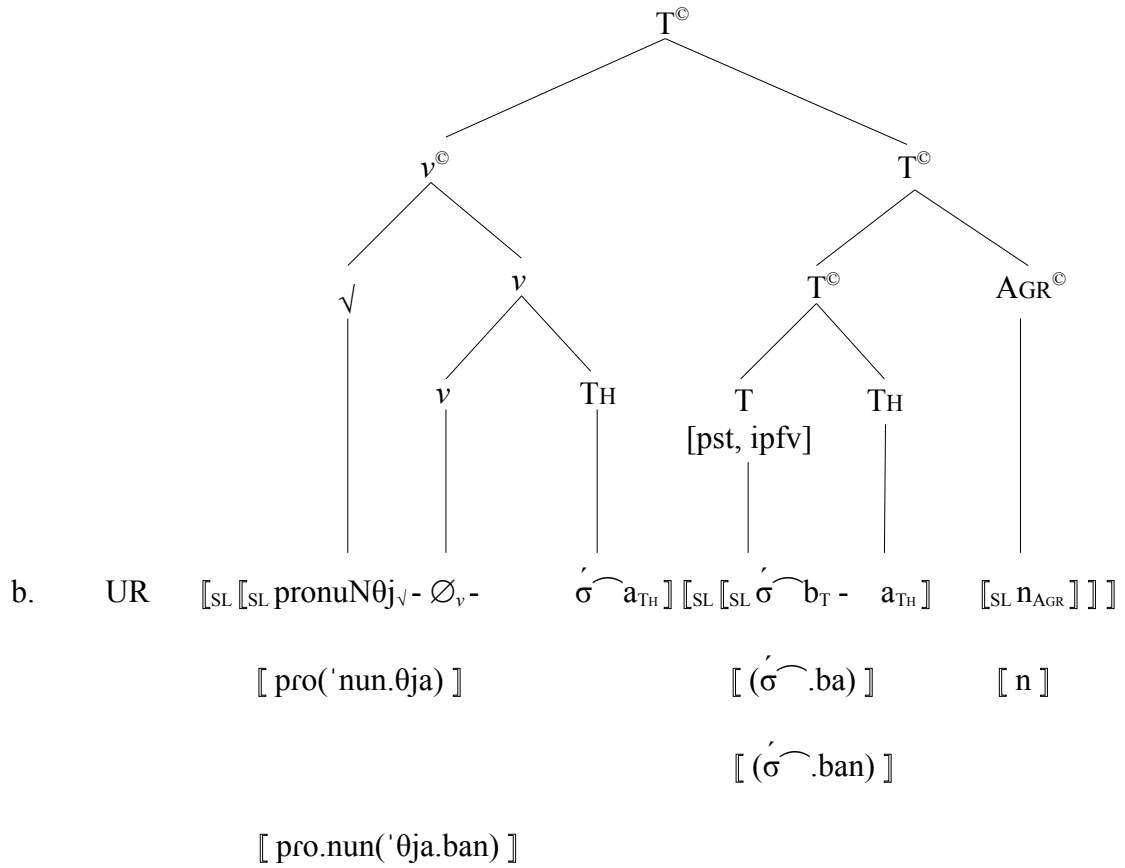
As mentioned in section 1, Cole (1991) claimed that there is no evidence at all for recursive stem-level domains in Spanish, whilst Myler (2015) asserted that only nominal and adjectival stems, but not verbal stems, trigger stem-level cycles in the language. These claims, however, are refuted by the cyclic transmission of high vocoid syllabification. Absent cyclic effects, the stem-level phonology of Spanish does not tolerate hiatus between an unstressed high vocoid and a following nonhigh vowel unless the high vocoid occupies the initial syllable and the nonhigh vowel is stressed (15) (Hualde 1997, 1999).



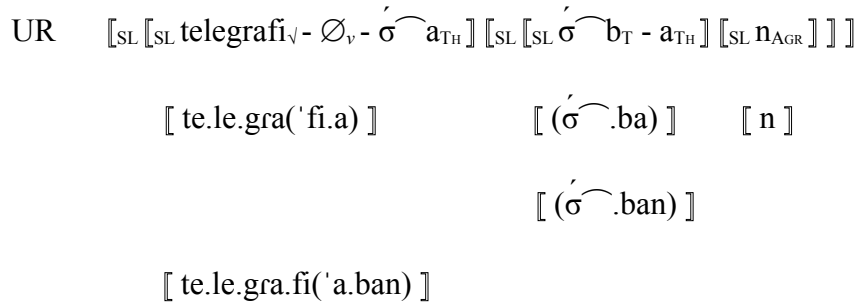
| (15) |                 |                |            | /I/ in first syllable? | Following V stressed? |
|------|-----------------|----------------|------------|------------------------|-----------------------|
| a.   | <i>piáno</i>    | [pi. 'a.no]    | ‘piano’    | yes                    | yes                   |
| b.   | <i>diagonál</i> | [dja.ɣo. 'nal] | ‘diagonal’ | yes                    | no                    |
| c.   | <i>religión</i> | [re.li. 'xjon] | ‘religion’ | no                     | yes                   |

The only exception occurs when the high vocoid bears stress in the immediately preceding cycle: in that case, hiatus is also tolerated (Navarro Tomás 1918:124; Bermúdez-Otero 2013:68-71, 2016:408-413), especially in Peninsular as opposed to Latin American varieties (Hualde et al. 2008:1908). This can be observed in the contrast between *pronunciában* [pro.nun. 'θja.βan] ‘pronounce.PST.IPFV.3PL’ and *telegrafiában* [te.le.ɣra.fi. 'a.βan] ‘telegraph.PST.IPFV.3PL’: the hiatus in the latter is licensed by the stress on /i/ in the cycle triggered by the stem *telegrafía* [te.le.ɣra. 'fi.a]. The full derivation is shown in (16), where we assume without modification the word-syntax motivated by Oltra-Massuet and Arregi (2005:50).

(16) a. *pronunciában* [pro.nun.'θja.βan] 'pronounce.PST.IPFV.3PL'



c. *telegrafiában* [te.le.gra.fi.'a.βan] 'telegraph.PST.IPFV.3PL'



These data, then, support principle (12b) as being fully general. Indeed we believe that the procedure described in (12) is likely to be universal.

### 3.2 Affixal cycles

The hypothesis that each affix undergoes a separate stem-level cycle predicts that affixes will exhibit the same metrical behavior as monomorphemic stems, except for the fact that  $\text{LEX} \leftrightarrow \text{PWD}$  (8a) prevents purely functional material from projecting a prosodic word. Depending on their size, therefore, different affixes may fail to project any suprasegmental structure at all, or may project up to the level of the syllable or the foot. Projection is driven by the constraints  $\text{PARSE-Seg}$  (17a) and  $\text{PARSE-}\sigma$  (17b) (Prince and Smolensky [1993] 2004), but blocked where appropriate by higher-ranked well-formedness requirements on syllable and foot structure.

- (17) a.  $\text{PARSE-Seg}$ : Every segment is dominated by a syllable node.  
 b.  $\text{PARSE-}\sigma$ : Every syllable is dominated by a foot node.  
 c.  $\text{PARSE-Ft}$ : Every foot is dominated by a prosodic word node.

Most affixes that are large enough to support a foot exhibit the default stress pattern. As shown in (18), for example, masculine diminutive *-it-o* ['i.to] projects a disyllabic bimoraic trochee, and only fails to project a prosodic word because of the superordinate status of  $\text{LEX} \leftrightarrow \text{PWD}$ .

- (18) *-it-o* ['i.to] 'DIM'

| <i>-it-o</i>                                                            | $\text{LEX} \leftrightarrow \text{PWD}$ | $\text{PARSE-}\sigma$ | $\text{PARSE-Ft}$ |
|-------------------------------------------------------------------------|-----------------------------------------|-----------------------|-------------------|
| $\rightarrow (\text{Ft} (\sigma \text{'i}^\mu) (\sigma \text{to}^\mu))$ |                                         |                       | *                 |
| $(\sigma \text{i}^\mu) (\sigma \text{to}^\mu)$                          |                                         | *W*                   | L                 |
| $(\omega (\text{Ft} (\sigma \text{'i}^\mu) (\sigma \text{to}^\mu)))$    | *W                                      |                       | L                 |

In turn, *-ez-Ø* ['eθ], which is used to derive the singular stem of deadjectival abstract nouns, instantiates default final stress on a heavy ultima (19a). This suffix belongs to the ordinary *e*-stem class and selects the null allomorph of its thematic element, like *caracól-Ø* in (6f). It contrasts with

pejorative *-ot-e* ['o.te], which, like *tomát-e* in (6f), surfaces with the *-e* allomorph of the theme because [t] is dispreferred domain-finally (19b). In both cases foot construction interacts transparently with allomorph selection and results in a right-aligned bimoraic trochee.

(19) a. *-ez-Ø* ['eθ] ‘-ness’

| <i>-eθ-</i> { $\emptyset$ ,e}                                                        | LEX $\leftrightarrow$ PWD | *t] | *ě] | PARSE- $\sigma$ |
|--------------------------------------------------------------------------------------|---------------------------|-----|-----|-----------------|
| $\rightarrow ({}_{\text{Ft}} ({}_{\sigma} 'e^{\mu}\theta^{\mu}))$                    |                           |     |     |                 |
| $({}_{\sigma} e^{\mu}\theta^{\mu})$                                                  |                           |     |     | *W              |
| $({}_{\omega} ({}_{\text{Ft}} ({}_{\sigma} 'e^{\mu}\theta^{\mu})))$                  | *W                        |     |     |                 |
| $({}_{\text{Ft}} ({}_{\sigma} 'e^{\mu}) ({}_{\sigma} \theta e^{\mu}))$               |                           |     | *W  |                 |
| $({}_{\sigma} e^{\mu}) ({}_{\sigma} \theta e^{\mu})$                                 |                           |     | *W  | *W*             |
| $({}_{\omega} ({}_{\text{Ft}} ({}_{\sigma} 'e^{\mu}) ({}_{\sigma} \theta e^{\mu})))$ | *W                        |     | *W  |                 |

b. *-ot-e* ['o.te] ‘PEJ’

| <i>-ot-</i> { $\emptyset$ ,e}                                                  | LEX $\leftrightarrow$ PWD | *t] | *ě] | PARSE- $\sigma$ |
|--------------------------------------------------------------------------------|---------------------------|-----|-----|-----------------|
| $({}_{\text{Ft}} ({}_{\sigma} 'o^{\mu}t^{\mu}))$                               |                           | *W  | L   |                 |
| $({}_{\sigma} o^{\mu}t^{\mu})$                                                 |                           | *W  | L   | *W              |
| $({}_{\omega} ({}_{\text{Ft}} ({}_{\sigma} 'o^{\mu}t^{\mu})))$                 | *W                        | *W  | L   |                 |
| $\rightarrow ({}_{\text{Ft}} ({}_{\sigma} 'o^{\mu}) ({}_{\sigma} te^{\mu}))$   |                           |     | *   |                 |
| $({}_{\sigma} o^{\mu}) ({}_{\sigma} te^{\mu})$                                 |                           |     | *   | *W*             |
| $({}_{\omega} ({}_{\text{Ft}} ({}_{\sigma} 'o^{\mu}) ({}_{\sigma} te^{\mu})))$ | *W                        |     | *   |                 |

There are affixes, however, which are not large enough to support a foot. These include a set of CV markers comprising, among others, first-person plural *-mo* /mo/ and several pronominal clitics such as third-person accusative *l-o* [lo], *l-a* [la], and dative *l-e* [le]. FTBIN( $\mu$ ) prevents these markers from projecting prosodic structure above the level of the syllable (20).

(20) *-mo* /mo/ ‘1PL’

| -mo                                                   | LEX $\leftrightarrow$ PWD | FTBIN( $\mu$ ) | PARSE-Seg | PARSE- $\sigma$ | PARSE-Ft |
|-------------------------------------------------------|---------------------------|----------------|-----------|-----------------|----------|
| $\rightarrow (\sigma \text{ mo}^\mu)$                 |                           |                |           | *               |          |
| mo                                                    |                           |                | *W*       | L               |          |
| $(_{\text{Ft}} (\sigma \text{ 'mo}^\mu))$             |                           | *W             |           | L               | *W       |
| $(_{\omega} (_{\text{Ft}} (\sigma \text{ 'mo}^\mu)))$ | *W                        | *W             |           | L               |          |

Incidentally, we know that the sequence *-mo-s* consists of two pieces, the first expounding first-person in the context of [plural] (32a,ii), the second discharging the feature [plural] itself (21c), because the combination of 1PL subject agreement with a 1PL reflexive object is realized as *-mo-Ø=no-s*, with only one [plural] marker: for example, *compré-mo-Ø=no-s=l-o* [kom.'pre.mo.nos.lo] ‘let us buy it for ourselves’. This is an instance of morphological haplogy or repetition avoidance (Nevins 2013, Cabré and Ohannesian 2020). The same effect causes 2PL imperative to be realized as */-Ø/* rather than */-d/* in the context of a 2PL reflexive object expounded by *=o-s*: for example, *comprá-d* [kom.'prað] ‘buy-IMP.2PL’, *comprá-Ø=o-s=l-o* [kom.'pra.os.lo] ‘buy it for yourselves!’

Finally, there is a small set of affixes consisting of a single consonant. Notably, *-s* [s] performs several functions (21a-c), and *-n* [n] realizes AGR[3pl] (21d).

- (21) a. NUM[pl]  $\leftrightarrow$  /-s/      e.g., *cára-s*      ['ka.ras]      ‘face-PL’  
b. AGR[2]  $\leftrightarrow$  /-s/      e.g., *cánta-s*      ['kan.tas]      ‘sing-2SG’  
c. AGR[pl]  $\leftrightarrow$  /-s/      e.g., *cánta-mo-s*      [kan.'ta.mos]      ‘sing-1(pl)-PL’  
d. AGR[3pl]  $\leftrightarrow$  /-n/      e.g., *cánta-n*      ['kan.tan]      ‘sing-3PL’

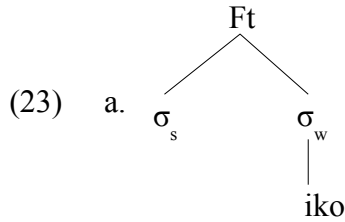
Single-consonant suffixes do not project any suprasegmental structure at all, but they are protected from stray erasure by the undominated faithfulness constraint MAX-Seg, which bans deletion. Of course, when these suffixes combine with a stem in a subsequent cycle, they are incorporated into syllable structure under pressure from PARSE-Seg. Crucially, the ranking MAX-Seg  $\gg$  PARSE-Seg is

only reversed at the word-level: it is only at that point that any material that may remain prosodically unaffiliated undergoes stray erasure. The same mechanism produces more spectacular effects in the case of French liaison, where consonants can remain floating at the word level and stray erasure is delayed until the phrase level (Bermúdez-Otero 2018a:188-195, Paschen 2018:36-37).

In sum, affixes containing at least two morae show the same default behavior as stems, projecting a bimoraic trochee. The default pattern of smaller affixes, in contrast, depends on the presence or absence of a vowel: if a vowel is present, the affix projects up to the syllable; if not, it does not project at all. Nonetheless, because the same stem-level constraint ranking holds in cycles triggered by affixes as in cycles triggered by minimal stems, the same nondefault patterns are possible in both. Just as an underlying prespecification gives rise to exceptional oxytonic stress in vowel-final *canapé* (7b), so we encounter affixes whose final vowel is prespecified as stressed: for example, demonym-forming *-í* ['i] (22a). Similarly, just as an underlying foot-head gives rise to exceptional proparoxytonic stress in *régimen* (10), so derivational *-ólisis* ['o.li.sis] is prespecified as stressed on its initial syllable (22b). In both cases, the underlying prespecification is protected by IDENT(stress) and satisfies \*LONG-LAPSE.

- |      |    |                      |                     |                |
|------|----|----------------------|---------------------|----------------|
| (22) | a. | <i>abas-í</i>        | [a.βa.'si]          | ‘Abbasid’      |
|      |    | <i>edris-í</i>       | [e.ðri.'si]         | ‘Idrisid’      |
|      |    | <i>guaran-í</i>      | [gwa.ra.'ni]        | ‘Guarani’      |
|      | b. | <i>electr-ólisis</i> | [e.lek.'tro.li.sis] | ‘electrolysis’ |
|      |    | <i>hem-ólisis</i>    | [e.'mo.li.sis]      | ‘haemolysis’   |
|      |    | <i>hidr-ólisis</i>   | [i.'ðro.li.sis]     | ‘hydrolysis’   |

Only one type of prespecification can be observed in affixes but not in minimal stems: preaccenting behavior. This occurs, for example, in adjectival *-ic-o* [i.ko], cognate with English *-ic* (e.g., *empír-ico* ‘empirical’). In this case, we assume that the initial vowel of the suffix is underlyingly specified as heading the weak branch of a disyllabic foot whose strong branch is empty: / $\acute{\sigma}$ iko/ (23a). This specification is protected by the faithfulness constraint IDENT- $\acute{\sigma}$  (23b). Like IDENT(stress), IDENT- $\acute{\sigma}$  is dominated by \*LONG-LAPSE, ensuring that, just as in (11b), a hypothetical preaccenting trisyllabic suffix such as / $\acute{\sigma}$ olisis/ will revert to the default pattern (23c).



b. IDENT- $\acute{\sigma}$ : If an input segment heads the weak branch of a disyllabic foot, so does its output correspondent.

c. hypothetical / $\acute{\sigma}$ olisis/

| $\acute{\sigma}$ olisis                    | *LONGLAPSE | IDENT- $\acute{\sigma}$ | ALIGN-R(Ft,D) |
|--------------------------------------------|------------|-------------------------|---------------|
| → o.li('si <sup>u</sup> s <sup>u</sup> )   |            | *                       |               |
| ('o <sup>u</sup> .li <sup>u</sup> )sis     |            | *                       | *W            |
| ( $\acute{\sigma}$ .o <sup>u</sup> )li.sis | *W         |                         | *W            |

This grammar allows in principle for the possibility that a minimal stem may be prespecified as preaccenting. However, multiple factors conspire to make such a possibility unlearnable. First, such a hypothetical specification would never survive in verbs, where accentual contrasts in stems are neutralized by the metrical properties of functional material (see section 3.3). Among nouns, \*LONG-LAPSE would restrict it to mono- or disyllabic stems. Even among those, in turn, preaccenting behavior would become manifest only under derivational prefixation, which is relatively infrequent.

This obviates the need for a grammatical restriction. If one is desired, however, it is trivially easy to implement: it suffices to demand that, within a prosodic word, every leftmost suprasegmental node must dominate the leftmost segment. This constraint prevents the left edge of a stem from bearing the metrical structure shown in (23a) because a stem must project a prosodic word (8a). In contrast, the constraint is vacuously satisfied by all affixes, even if specified as in (23a), because affixes do not project prosodic words.

In sum, the grammar that describes the metrical behavior of minimal stems (section 3.1) captures, subject to only minimal additions, the metrical properties of affixes.

### *3.3 Complex items, with special attention to inflected verbs*

We are now ready to address the metrification of complex items, beginning with those created by means of stem-level affixation.

We first need an operational criterion for identifying stem-level suffixes. In Spanish such a criterion is found in the diphthongal alternation (Halle, Harris and Vergnaud 1991:146; Bermúdez-Otero 2006:286; 2013:section 3). This is an instance of allomorphy manifested by an idiosyncratic set of bases, which surface sometimes with a stressed diphthong (i.e., ['je] or ['we]), sometimes with an unstressed monophthong (i.e., [e] or [o]). Now take a base which, in isolation, surfaces with a stressed diphthong (24a), and add a suffix to it. That suffix will be diagnosed as belonging to the stem level if it is the case that, when its presence induces stress shift, the base appears in its monophthongal allomorph (24b). By this criterion, the set of stem-level suffixes of Spanish includes most derivational suffixes, as well as verbal inflection up to, and including, AGR. Suffixes that do not meet the criterion are identified as word-level. This comprises suffixes that cause stress shift but induce the opaque selection of the diphthongal allomorph (24c), alongside suffixes that never cause stress shift (24d). The resulting class of word-level suffixes includes evaluative markers in denominal and deadjectival (but not deverbal) use, the superlative *-ísimo* ['i.si.mo], the circumfix



$a\text{-...-}\emptyset_v\text{-}a_{\text{Th}}$  (which creates deadjectival change-of-state verbs), and pronominal enclitics.

|      |    |                     |                   |                              |
|------|----|---------------------|-------------------|------------------------------|
| (24) | a. | <i>pliéga</i>       | [ 'plje.ɣa]       | ‘fold (verb)’                |
|      |    | <i>pliégue</i>      | [ 'plje.ɣe]       | ‘fold (noun)’                |
|      |    | <i>viéjo</i>        | [ 'bje.xo]        | ‘old’                        |
|      | b. | <i>plega-dór</i>    | [ple.ɣa. 'ðor]    | ‘one that folds’             |
|      |    | <i>plegá-ba</i>     | [ple. 'ɣa.βa]     | ‘fold-PST.IPFV.1/3SG’        |
|      |    | <i>plegá-mo-s</i>   | [ple. 'ɣa.mos]    | ‘fold-1-PL’                  |
|      | c. | <i>viej-ón</i>      | [bje. 'xon]       | ‘old-AUG’                    |
|      |    | <i>viej-ísimo</i>   | [bje. 'xi.si.mo]  | ‘old-SUPL’                   |
|      |    | <i>a-viej-á-r</i>   | [a.βje. 'xar]     | ‘to cause to become old’     |
|      | d. | <i>pliéga=me=lo</i> | [ 'plje.ɣa.me.lo] | ‘fold.IMP=1SG.DAT=3SG.M.ACC’ |
|      |    |                     |                   |                              |

Now, since every minimal stem and every affix has already been independently metrified in an initial cycle of the stem-level phonology (12a), the computation of the stress profile of complex items reduces to an extremely simple generalization: the rightmost input foot is preserved and all other feet are deleted. This is illustrated in (25); note that, in this example, a regular phonological process removes the final vowel of the stem before the initial vowel of the suffix (Bermúdez-Otero 2013:33-50).

(25) *belléza* [be. 'ʎe.θa] ‘beauty’

|    |                                                                                                                                              |              |                     |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------|
|    |                                                                                                                                              | ‘beautiful’  | ‘-ness’             |
| UR | [ <sub>SL</sub> [ <sub>SL</sub> beʎ <sub>v</sub> - $\emptyset_a$ - o <sub>Th</sub> ] [ <sub>SL</sub> eθ <sub>n</sub> - a <sub>Th</sub> ] ] ] |              |                     |
|    | [ ( 'be.ʎo) ]                                                                                                                                | [ ( 'e.θa) ] |                     |
|    | [ be( 'ʎe.θa) ]                                                                                                                              |              | cf. *[ ( 'be.ʎe)θa] |

The deletion of input feet is driven by the markedness constraint \*W<sub>KFT</sub>, which has the effect of allowing no more than one foot to survive (26a). \*W<sub>KFT</sub> dominates the faithfulness constraints IDENT(stress) (7a) and IDENT- $\acute{\sigma}$  (23b), as well as PARSE- $\sigma$  (17b). Deciding which foot survives is down to the faithfulness constraint RIGHTMOSTWINS, which protects the rightmost input foot (26b).

(26) a. \*W<sub>KFT</sub>: Assign one violation mark for each weak foot.

b. RIGHTMOSTWINS: Let *s* be the rightmost input segment that heads a footed syllable: if *s* heads a strong foot-branch, so does its output correspondent; if *s* heads a weak foot-branch, so does its output correspondent.

c. *belléz-a* [be. 'ʎe.θa] ‘beauty’

| $(_{\omega} (_{Ft} 'be.ʎo))-(_{Ft} 'e.θa)$                 | RIGHTMOSTWINS | *W <sub>KFT</sub> | IDENT(stress) | PARSE- $\sigma$ |
|------------------------------------------------------------|---------------|-------------------|---------------|-----------------|
| $\rightarrow (_{\omega} be(_{Ft} 'ʎe^{\mu}.θa^{\mu}))$     |               |                   | *             | *               |
| $(_{\omega} (_{Ft} , be^{\mu})(_{Ft} 'ʎe^{\mu}.θa^{\mu}))$ |               | *W                | L             | L               |
| $(_{\omega} (_{Ft} 'be^{\mu}.ʎe^{\mu})θa)$                 | *W            |                   | *             | *               |

Of course, the final trisyllabic window remains in operation throughout all stem-level cycles: accordingly, \*LONGLAPSE is still undominated (11b, 23c).

The mechanisms motivated thus far suffice to capture all the stress properties of inflected verbs, which in previous analyses have often been treated partially or wholly separately: notably, Harris set verbs aside in his classical 1983 study (see, e.g., p. 84), whilst more recently Oltra-Massuet and Arregi (2005) focus on verbs and only briefly discuss other categories. Our proposal, in contrast, subsumes the stress behavior of verbs under the single system described above. In particular, we will now show how the grammar derives (i) the neutralization of root stress contrasts in verb stems and (ii) the well-known phenomenon of fixed or “columnar” stress within the paradigms of individual tenses.

It is a long-established observation that stress is always fully predictable throughout the paradigm of a Spanish regular verb (see again Harris 1983:84). In our analysis, this is because phonemic stress distinctions among roots are neutralized in the phonological cycles triggered by minimal verb stems. This neutralization is caused by the metrical prespecification of the theme vowel adjoined to the *v* head, which happens to be preaccenting. In (27a), for example, we see that the bound root shared by the noun *súplica* and the verb *suplica* is underlyingly prespecified as stressed on the initial syllable; it thus contrasts with the roots of phonologically similar nouns like *botíca* and *caníca*, which exhibit default stress (27b). In the verb *suplica*, however, the underlying foot-head of the root fails to survive (27c).

- (27) a. *súplica* [ˈsu.pli.ka] ‘supplication’ (< Latin *súpplicō* < *sub-plīcō*)  
*fábrica* [ˈfa.βri.ka] ‘factory’ (< Latin *fābrīca* < *fab(e)r-īcus*)
- b. *botíca* [bo.ˈti.ka] ‘pharmacy’ (< French *boutique* < Greek ἀποθήκη)  
*caníca* [ka.ˈni.ka] ‘marble (toy)’ (< French *canique* < Dutch *knikker*)
- c. *suplica* [su.ˈpli.ka] ‘supplicate’ (cf. Italian *lui súpplica*)  
*fabríca* [fa.ˈβri.ka] ‘manufacture’ (cf. Italian *lui fábbrica*)

As noted, the divergent behavior of the noun *súplica* and the verb *suplica* is caused by the underlying distinction between their respective theme vowels: the  $/-a_{TH}/$  adjoined to *n* is metrically unspecified (28a), whereas the  $/-\acute{\sigma}a_{TH}/$  adjoined to *v* is preaccenting (28b).

- (28) a. *súplica* ‘supplication’  
 $[[_{SL} \acute{s}uplik_{\check{v}} - \emptyset_n - a_{TH}]]$   
 (ˈsu.pli)ka
- b. *suplica* ‘supplicate’  
 $[[_{SL} \acute{s}uplik_{\check{v}} - \emptyset_v - \acute{\sigma}a_{TH}]]$   
 su(ˈpli.ka)

In section 4.2 we will show that this analysis misses no generalizations and complies in full with Richness of the Base. In particular, even if one limits oneself to the verbal domain, one can find theme vowels of all the permitted accentual types: unaccented, accented, and preaccenting.

We now use the example of the past imperfective to illustrate the emergence of columnar stress within individual tense paradigms (29). As per (16), we adopt without modification the word-syntax motivated by Oltra-Massuet and Arregi (2005:50), with theme vowels identified in accordance with Oltra-Massuet 1999.

|      |                       |   |     |   |     |      |                     |
|------|-----------------------|---|-----|---|-----|------|---------------------|
| (29) | ‘supplicate.PST.IPFV’ |   |     |   |     |      |                     |
|      | √                     | v | TH  | T | TH  | AGR  |                     |
| 1SG  | súplik                | ∅ | σ̇a | b | σ̇a | ∅    | [su.pli('ka.βa)]    |
| 2SG  | súplik                | ∅ | σ̇a | b | σ̇a | s    | [su.pli('ka.βas)]   |
| 3SG  | súplik                | ∅ | σ̇a | b | σ̇a | ∅    | [su.pli('ka.βa)]    |
| 1PL  | súplik                | ∅ | σ̇a | b | σ̇a | mo-s | [su.pli('ka.βa)mos] |
| 2PL  | súplik                | ∅ | σ̇a | b | σ̇a | i-s  | [su.pli('ka.βajs)]  |
| 3PL  | súplik                | ∅ | σ̇a | b | σ̇a | n    | [su.pli('ka.βan)]   |

Under our principles of cyclic domain assignment (12a,b), this syntactic structure defines minimal stem-level cyclic domains over the stem and over each suffix. Recall that theme vowels, where present, belong in the same cyclic domain as the piece they are adjoined to because TH, like √, is not a projecting syntactic category: see (13) again.

In (30) we illustrate the course of the derivation with the case of *suplicábamos* ‘supplicate.PST.IPFV.1PL’, where, strikingly, columnar stress results in a highly marked proparoxytonic contour with a heavy ultima: see (10c). We start with the smallest cyclic domains.

In the cycle triggered by the stem, the preaccenting specification of the theme vowel, being closer to the right edge, wins over the foot-head on the initial syllable of the root; this yields [su('pli.ka)], see (26c). The tense marker becomes preaccenting by retaining the underlying metrical structure associated with its theme vowel: [(σ<sup>ˈ</sup>.ba)], see (23). In turn, both AGR exponents fail to project feet because they are too small. *-Mo-*, which realizes first person, projects up to the syllable: [.mo.], see (20). Finally, *-s*, which realizes plural (21c), does not project any suprasegmental structure: [s]. In the final stem-level cycle, the last syllable of the stem fills the head of the input foot whose weak branch is occupied by the tense marker; this foot survives because it is the rightmost, whereas the stress on the second syllable of the stem is deleted: [su.pli('ka.βa)mos], see (26c).

(30) *suplicábamos* [su.pli. 'ka.βa.mos] ‘supplicate.PST.IPFV.1 PL’

UR                      [SL [SL súplik<sub>v</sub> - Ø<sub>v</sub> - σ<sup>ˈ</sup>a<sub>Th</sub>] [SL [SL σ<sup>ˈ</sup>b<sub>T</sub> - a<sub>Th</sub>] [SL mo<sub>AGR</sub>] [SL s<sub>AGR</sub>] ] ] ]  
                                  [ su('pli.ka) ]                      [ (σ<sup>ˈ</sup>.ba) ]                      [ .mo. ]                      [ s ]  
                                                                                  [ (σ<sup>ˈ</sup>.ba)mos ]  
                                  [ su.pli('ka.βa)mos ]

This analysis generalizes to all cases of columnar stress, subject only to the different underlying specifications of the relevant T exponents: we shall see in section 4.2, for example, that the preterite of second- and third-conjugation verbs is realized by a zero morph /Ø<sub>T</sub>/ with an adjoined accented theme vowel /-í<sub>Th</sub>/ (37b).

Only one tense fails to exhibit columnar stress: the present tense. We give its syntactic structure in (31). Again following Oltra-Massuet and Arregi (2005:59), we assume that T[prs] fuses with AGR.

|      |                      |   |                  |                     |                   |
|------|----------------------|---|------------------|---------------------|-------------------|
| (31) | ‘supplicate.PRS.IND’ |   |                  |                     |                   |
|      | √                    | v | T <sub>H</sub>   | T/AGR               |                   |
| 1SG  | súplik               | ∅ | σ <sup>ˈ</sup> a | o                   | [su(ˈpli.ko)]     |
| 2SG  | súplik               | ∅ | σ <sup>ˈ</sup> a | s                   | [su(ˈpli.kas)]    |
| 3SG  | súplik               | ∅ | σ <sup>ˈ</sup> a | ∅                   | [su(ˈpli.ka)]     |
| 1PL  | súplik               | ∅ | σ <sup>ˈ</sup> a | σ <sup>ˈ</sup> mo-s | [su.pli(ˈka.mos)] |
| 2PL  | súplik               | ∅ | σ <sup>ˈ</sup> a | σ <sup>ˈ</sup> i-s  | [su.pli(ˈkajs)]   |
| 3PL  | súplik               | ∅ | σ <sup>ˈ</sup> a | n                   | [su(ˈpli.kan)]    |

The forms that require special treatment are first-person plural [su.pli(ˈka.mos)] and second-person plural [su.pli(ˈkajs)]. These fail to preserve the stress assigned to the second syllable of the stem [su(ˈpli.ka)] during the first cycle. Since ungrammatical \*[su(ˈpli.ka)mos] and \*[su(ˈpli.kajs)] would both have satisfied the final three-syllable window, cf. (9b, 10c, 30), this phenomenon indicates that, in these forms, stem stress loses out to an input foot-head located further to the right. We must therefore conclude that the first- and second-person plural markers exhibit allomorphy, being realized by preaccenting exponents in present-tense contexts: /-σ<sup>ˈ</sup>mo/, /-σ<sup>ˈ</sup>i/ (32).

- (32) a. i. AGR[1]  $\leftrightarrow$  /-σ̣mo/ / [prs] \_\_ [pl]  
 ii. AGR[1]  $\leftrightarrow$  /-mo/ / \_\_ [pl]  
 b. *suplicámos* [su.pli.'ka.mos] ‘supplicate.PRS.1PL’  
 UR  $\llbracket_{SL} \llbracket_{SL} \text{súplik}_\vee - \emptyset_\vee - \sigmạ \text{a}_{TH} \rrbracket \llbracket_{SL} \llbracket_{SL} \sigmạ \text{mo}_{AGR} \rrbracket \llbracket_{SL} \text{s}_{AGR} \rrbracket \rrbracket$   
 $\llbracket \text{su.('pli.ka)} \rrbracket \quad \llbracket (\sigmạ \text{.mo)} \rrbracket \quad \llbracket \text{s} \rrbracket$   
 $\llbracket (\sigmạ \text{.mos)} \rrbracket$   
 $\llbracket \text{su.pli('ka.mos)} \rrbracket$

In our analysis, the unaccented exponents of the first- and second-person plural markers are the elsewhere allomorphs. This predicts that they will be selected by default unless blocked by the preaccenting allomorphs, which have a more specific distribution. As we shall see in section 5, this prediction is strongly confirmed by dialects in which preaccenting /-σ̣mo/ is no longer available to present subjunctive forms, whose frequency in discourse is lower than that of the present indicative. We thus correctly derive the generalization that columnar verb stress expands diachronically at the expense of alternating stress and that, in the case of the present, it leads to stress on the last vowel of the root.

### 3.4 The word level

Setting compounds aside, each word-level derivative consists of a base plus one or more word-level affixes. Each of these pieces has gone through its own individual stem-level cycle. Together, they now undergo a single pass through the word-level phonology because word-level domains are not recursive (12c). In Spanish, the ranking of metrical constraints is the same in both lexical strata, with just one exception: \*LONGLAPSE (11a) is demoted below IDENT(stress) (7a) and IDENTσ̣̣\_\_ (23b) at the word level. In that stratum, therefore, stress continues to fall on the rightmost syllable

specified as a foot-head in the input, but is now no longer confined within the domain-final trisyllabic window.

In this system, a word-level derivative can come to exhibit a nondefault stress pattern in one of two ways. First, its rightmost accented constituent may itself happen to bear a nondefault specification: this is the case of superlative *-ísimo* (33).

(33) *bellísimo* [be.'ʎi.si.mo] ‘most beautiful’

|    |                                                                                                                                                                  |                                        |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|    | ‘beautiful’                                                                                                                                                      | ‘SUPL’                                 |
| UR | $[_{\text{WL}} [_{\text{SL}} \text{be}\lambda_{\text{V}} - \emptyset_a - \text{o}_{\text{TH}}] [_{\text{SL}} \text{ísim}_{\text{SUPL}} - \text{o}_{\text{TH}}]]$ |                                        |
|    | $[ (' \text{be}.\lambda\text{o}) ]$                                                                                                                              | $[ (' \text{i}.\text{si})\text{mo}) ]$ |
|    | $[ \text{be}(' \lambda\text{i}.\text{si})\text{mo} ]$                                                                                                            |                                        |

Alternatively, one or more of the suffixes following the base may be unaccented owing to their small size, so that base stress is preserved. In the case of plural *sábana-s* (34), for example, the suffix /-s/ is too small to project suprasegmental structure in its stem-level cycle and, when concatenated with the base, it creates a highly marked proparoxytonic contour with a heavy ultima: see (10c). There is thus no need to specify the plural suffix as extrametrical; its prosodic behavior is exhaustively explained by its size, given independently motivated cyclic domain structure.

(34) *sábanas* ['sa.βa.nas] ‘bedsheets’

|    |                                                                                                                                       |                |
|----|---------------------------------------------------------------------------------------------------------------------------------------|----------------|
|    | ‘bedsheet’                                                                                                                            | ‘PL’           |
| UR | $[_{\text{WL}} [_{\text{SL}} \text{sában}_{\text{V}} - \emptyset_n - \text{a}_{\text{TH}}] [_{\text{SL}} \text{s}_{\text{NUM[pl]}}]]$ |                |
|    | $[ (' \text{sa}.\text{ba})\text{na} ]$                                                                                                | $[ \text{s} ]$ |
|    | $[ (' \text{sa}.\text{ba})\text{nas} ]$                                                                                               |                |



A more spectacular example is provided by pronominal enclitics, which trigger violations of the final trisyllabic window (35). Once more, no special stipulation is needed beyond the demotion of \*LONGLAPSE in the word-level constraint hierarchy: the clitics' small size explains their unaccentedness, and their unaccentedness in turn explains the preservation of the metrical structure of the base.

(35) *cántamelo* ['kan.ta.me.lo] 'sing it to me!'

|    |                                                                                                                                                                                                                                                                                                                              |                                     |                                     |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
|    | ‘sing.IMP.SG’                                                                                                                                                                                                                                                                                                                | ‘1SG.DAT’                           | ‘3ACC’                              |
| UR | $\llbracket_{\text{WL}} \llbracket_{\text{SL}} \text{kant}_{\text{V}} - \emptyset_{\text{V}} - \acute{\sigma} \text{a}_{\text{TH}} \rrbracket \llbracket_{\text{SL}} \text{m}_{1\text{SG.DAT}} - \text{e}_{\text{TH}} \rrbracket \llbracket_{\text{SL}} \text{l}_{3\text{ACC}} - \text{o}_{\text{TH}} \rrbracket \rrbracket$ |                                     |                                     |
|    | $\llbracket (' \text{kan.ta} ) \rrbracket$                                                                                                                                                                                                                                                                                   | $\llbracket .\text{me.} \rrbracket$ | $\llbracket .\text{lo.} \rrbracket$ |
|    | $\llbracket (' \text{kan.ta})\text{me.lo} \rrbracket$                                                                                                                                                                                                                                                                        |                                     |                                     |

## 4 Theoretical foundations, empirical predictions

### 4.1 Modularity, cyclicity, constrained nonuniformity

The analysis we have just set out rests on a package of foundational assumptions comprising qualified principles of modularity, cyclicity, and constrained nonuniformity.

First, modularity renders the theory of grammar more restrictive by forbidding all but a narrow set of crossmodular statements. This enhanced restrictiveness, in turn, yields two advantages: improved learnability and increased empirical content (Bermúdez-Otero 2012:45-49). Notably, a sharp demarcation between morphology and phonology is needed to capture the basic observation that suppletive patterns are never systematically extended through the lexicon, even when they enjoy a relatively high degree of statistical confidence (Bermúdez-Otero 2018a). Our analysis accordingly upholds a high standard of modularity. In particular, no morphological or syntactic property is named in the statement of our phonological constraints, except for the broad distinction between stems and affixes. Indeed, the technical formulation of  $\text{LEX} \leftrightarrow \text{PWD}$  in (8a) refers

only to stem domains, defined as those comprising some nonfunctional material; it does not mention the predicate ‘affix’, thereby complying with the Lexical Category Condition (Truckenbrodt 1999:226, and references cited therein). The need for phonology to refer to the lexical-functional distinction is in any case entirely uncontroversial, as there is overwhelming crosslinguistic evidence for phonological asymmetries between stems and affixes (e.g., Gouskova 2023). We independently know that the phonology also needs to be able to distinguish configurations contained within a single input piece from those arising from the concatenation of two or more input pieces, where the size of input pieces varies across strata: at the stem level, input pieces consist of roots, stems, and affixes; at the phrase level, of grammatical words. The stratal theory of morphosyntactic colors (Trommer 2015) provides a highly restrictive encoding of this distinction by reducing it to the predicates ‘tautochromatic’ and ‘heterochromatic’, although even these happen not to be needed in the description of Spanish stress patterns. In contrast, conceptions of the interface that allow the phonology freer and more fine-grained access to morphosyntactic structure stand sharply at odds with the theoretical framework espoused in this article; in section 5 they will be shown to lead to overgeneralization in the analysis of Spanish stress, thus failing to vindicate their increased permissiveness.

This high standard of modularity is crucially underwritten by cyclicity: beyond the distinction between stems and affixes, morphosyntactic structure determines the course of stress assignment in Spanish solely by controlling the size and stratal affiliation of domains for phonological computation. As we have seen, this produces decisive effects: a salient example is that of affixes which, being too small, fail to project prosodic structure in their cycles, later behaving as stress-neutral when concatenated with a stem. In this connection, it is particularly important that, once individual affixes have been assigned to the stem or word level, cyclic domains are defined by a procedure that is completely general and, indeed, plausibly universal (12).

In turn, the analysis admits some degree of nonuniformity in the phonology, but kept within

strict theoretical control. As we noted in section 1, we countenance just two sources of nonuniformity: lexical prespecification and stratification. Lexical prespecification is directly controlled by the stem-level phonological grammar via the relative ranking of faithfulness and markedness constraints under Richness of the Base. This mechanism is altogether indispensable, and indeed the same that regulates phonemic contrast (Kager 2009). The second source of nonuniformity in our analysis comes from the contrast between the stem-level and word-level phonology. This reduces to a binary choice between two strata, each of which is associated with a single constraint ranking. Thus, nonuniformity is kept under control by Richness of the Base and by the requirement of a single constraint ranking per stratum. These principles generate strikingly successful empirical predictions, to which we now turn.

#### *4.2 Richness of the Base*

Under Richness of the Base, the grammar is responsible for enforcing all relevant linguistic generalizations without aid from morpheme structure constraints. The latter's traditional functions, as we have seen, are taken over by the stem-level phonology in interaction with other factors, notably the exponence of functional material (cf. Rasin and Katzir 2020, Tebay 2021, Rasin 2025). In Spanish, for example, the stem-level constraint hierarchy filters out all but a narrow range of prespecified stress patterns (section 2.2); this range is then further restricted in verbs because the theme vowels attaching to the *v* head happen to be preaccenting, thus masking the metrical properties of roots (27, 28). This approach generalizes to other cases, such as that of tone in Bari verbs, recently discussed by Trommer and Gleim (2025): unlike Bari nouns (Yokwe 1986:125), Bari verbs fall into just two tonal classes, H and LHL (Yokwe 1986:11-14), because the *v* head happens to have two circumfixal tonal exponents that overwrite root tones. But in general, if there are no morpheme structure constraints, we expect phonological properties to be unpredictably distributed over underlying representations (*modulo* truly accidental gaps). In our case, this predicts

that there will be no restrictions on the occurrence of the crucially active metrical prespecifications allowed by the stem-level phonology of Spanish: any of these specifications should be able to occur in any type of morph. This means, in turn, that the distinction between unspecified and crucially prespecified morphs should not coincide with any independently motivated distinction such as those between stems and affixes, derivation and inflection, or nouns and verbs. Gratifyingly, this is exactly what we find, despite initial appearances to the contrary in the case of verbs.

Recall that, in our analysis, the stem-level phonology of Spanish permits four types of crucial metrical prespecification: (i) a morph-final vowel may be specified as stressed, generating a marked oxytonic contour in its domain; (ii) a penultimate vowel in a morph may be prespecified as stressed before a heavy syllable, resulting in a marked paroxytonic contour; (iii) an antepenultimate syllable may also be specified as stressed, resulting in a marked proparoxytonic profile; and, finally, (iv) an affix may be specified as preaccenting. All but one of these prespecification types occur both in stems and in affixes; the exception is preaccentuation, which, for independent reasons discussed at the end of section 3.2, is either extremely improbable or absolutely impossible in stems. Thus, stems and affixes exhibit parallel behavior in respect of metrical prespecification: this is illustrated with examples of type (i) in (36a), type (ii) in (36b), and type (iii) in (36c,d).

| (36) | stem                 | affix                    |       |                      |                     |                  |
|------|----------------------|--------------------------|-------|----------------------|---------------------|------------------|
| a.   | <i>canapé</i> (7b)   | <i>-í</i> (22a)          | e.g., | <i>abas-í</i>        | [a.βa.'si]          | ‘Abbasid’        |
|      |                      |                          |       | <i>israel-í</i>      | [is.ra.e.'li]       | ‘Israeli’        |
| b.   | <i>azúcar</i> (9b)   | <i>-átil</i> ‘-atile’    | e.g., | <i>vers-átil</i>     | [ber.'sa.til]       | ‘versatile’      |
|      |                      |                          |       | <i>port-átil</i>     | [por.'ta.til]       | ‘portable’       |
| c.   | <i>sábana</i> (10c)  | <i>-ísimo</i> (33)       | e.g., | <i>fidel-ísimo</i>   | [fi.ðe.'li.si.mo]   | ‘most faithful’  |
|      |                      |                          |       | <i>bell-ísimo</i>    | [be.'li.si.mo]      | ‘most beautiful’ |
| d.   | <i>régimen</i> (10c) | <i>-ópolis</i> ‘-opolis’ | e.g., | <i>metr-ópolis</i>   | [me.'tro.po.lis]    | ‘metropolis’     |
|      |                      |                          |       | <i>gigant-ópolis</i> | [xi.ɣan.'to.po.lis] | ‘city of giants’ |

Theme vowels provide a particularly stringent test of this prediction. Our analysis might seem vulnerable to the objection that a systematic contrast between nominal and verbal stress has been surreptitiously smuggled into the derivation under the cover of the different metrical specifications of nominal and verbal theme vowels: recall that theme vowels adjoined to *n* or *a* are unaccented, whereas theme vowels adjoined to *v* are preaccenting. However, such an objection would be misguided. Closer inspection of the full inventory of theme vowels in the verbal domain reveals the existence of preaccenting, accented, and unaccented members. More specifically, preaccenting theme vowels in the verbal domain include not just those adjoined to *v* (37a), but also those adjoined to the T exponent in the past imperfective (16, 29, 30). In the preterite, in turn, the T<sub>H</sub> position adjoined to the T head is realized by a theme vowel that is underlyingly accented (37b). Finally, the conditional marker */-rí-/* is itself accented and followed by an unaccented theme vowel (37c).

(37) a. Preaccenting theme vowel

*suplicas* [su.'pli.kas] 'supplicate.2SG'

i.  $\sqrt{\quad} \quad \nu \quad \mathbf{TH} \quad \mathbf{T/AGR}$

súplik  $\emptyset$   $\acute{\sigma} \curvearrowright \mathbf{a}$  s

ii.  $[[_{\text{SL}} [[_{\text{SL}} \text{súplik}_{\sqrt{\quad}} - \emptyset_{\nu} - \acute{\sigma} \curvearrowright \mathbf{a}_{\mathbf{TH}}] ] [_{\text{SL}} \mathbf{S_{T/AGR}}] ] ]$

$[[ \text{su}(' \text{pli.k} \mathbf{a}) ] \quad [ \text{s} ]$

$[[ \text{su}(' \text{pli.kas}) ]$

b. Accented theme vowel

*temíste* [te.'mis.te] 'fear.PST.PRF.2SG'

i.  $\sqrt{\quad} \quad \nu \quad \mathbf{TH} \quad \mathbf{T} \quad \mathbf{Th} \quad \mathbf{AGR}$

tem  $\emptyset$   $\acute{\sigma} \curvearrowright \mathbf{e}$   $\emptyset$   $\mathbf{i}$  ste

ii.  $[[_{\text{SL}} [[_{\text{SL}} \text{tem}_{\sqrt{\quad}} - \emptyset_{\nu} - \acute{\sigma} \curvearrowright \mathbf{e}_{\mathbf{TH}}] ] [_{\text{SL}} [[_{\text{SL}} \emptyset_{\mathbf{T}} - \mathbf{i}_{\mathbf{TH}}] ] [_{\text{SL}} \text{ste}_{\mathbf{AGR}}] ] ] ]$

$[[ (' \text{te.me}) ] \quad [ (' \mathbf{i}) ] \quad [ \text{s.te.} ]$

$[[ (' \text{is.te}) ]$

$[[ \text{te}(' \text{mis.te}) ]$

c. Unaccented theme vowel

*vivirías* [bi.βi.'ri.as] 'live.COND.2SG'

i.  $\sqrt{\quad} \quad \nu \quad \mathbf{TH} \quad \mathbf{T} \quad \mathbf{Th} \quad \mathbf{AGR}$

bib  $\emptyset$   $\acute{\sigma} \curvearrowright \mathbf{i}$  rí  $\mathbf{a}$  s

ii.  $[[_{\text{SL}} [[_{\text{SL}} \text{bib}_{\sqrt{\quad}} - \emptyset_{\nu} - \acute{\sigma} \curvearrowright \mathbf{i}_{\mathbf{TH}}] ] [_{\text{SL}} [[_{\text{SL}} \mathbf{r} \acute{\mathbf{i}}_{\mathbf{T}} - \mathbf{a}_{\mathbf{TH}}] ] [_{\text{SL}} \mathbf{S_{AGR}}] ] ] ]$

$[[ (' \text{bi.bi}) ] \quad [ (' \text{ri.a}) ] \quad [ \text{s} ]$

$[[ (\text{rí.as}) ]$

$[[ \text{bi.bi}(' \text{ri.as}) ]$

In this light, theme vowels exhibit a greater variety of metrical behaviors in the verbal than in the nominal domain not because verbs and nouns have different phonological grammars, but rather because nominal theme vowels have a very restricted distribution, occurring only in adjunction to *n* or *a*, whereas verbal theme vowels occur in a far wider set of environments: namely, in adjunction to *v* or to several different flavours of *T*. As a result, there are just many more types of verbal theme vowels than of nominal theme vowels.

#### *4.3 Chung's Generalization and the Monomorpheme Generalization*

So far, we have seen that the distribution of underlying metrical prespecifications fully complies with the predictions entailed by Richness of the Base. But recall that there is a second source of nonuniformity: the binary choice between the stem-level and the word-level phonology. This too is under stringent theoretical control insofar as all domains affiliated to the same stratum are subject to the same ranking of phonological constraints; parallel cophonologies within the same stratum are forbidden (cf. Orgun 1996, Inkelas 1998, Sande et al. 2020). It so happens, however, that the set of stem-level items comprises highly diverse constructions: as illustrated in (2), stem-level items may be overtly suffixed or not, and may be monocyclic or polycyclic. In consequence, the prohibition of parallel cophonologies within a stratum yields a particularly strong prediction in the case of the stem level: despite considerable diversity among stem-level constructions, exactly the same range of metrical profiles is available to all of them.

We thus predict that monocyclic and polycyclic stem-level items are phonologically homogeneous. This idea has a long history, dating back to the 1980s: it is known in the literature as Chung's Generalization (38) (named after Chung 1983: 63; see Bermúdez-Otero and McMahon 2006:399-400; Kiparsky 2007:20-27; Bermúdez-Otero 2012:31-33, 2018:110-111, 2019:3-8). In (38) we provide a technical statement which highlights the status of the generalization as a theorem entailed by the principle that the same ranking of faithfulness and markedness constraints applies in

all stem-level cycles.

(38) *Chung's Generalization*

If a specification in a given environment in the rich base is faithfully preserved in the output of monocyclic stem-level derivations, then the same specification in the same environment is also faithfully preserved between cycles in polycyclic stem-level derivations, and vice versa.

In observational terms, (38) means that, if a property is contrastive in items defining a single stem-level domain (monomorphemes or root-based items), it must also be preserved across stem-level cycles.

The empirical evidence of Spanish stress assignment is in robust agreement with Chung's Generalization. Recall from section 3.3 that the diphthongization criterion enables us to draw the distinction between stem-level and word-level suffixes independently (24). Once this distinction is in place, it is easy to verify that those metrical patterns, whether default or marked, which occur in monocyclic stem-level items can also arise via cyclic preservation in polycyclic stem-level items. For example, *azúcar* exhibits marked paroxytonic stress before a heavy ultima (9), but exactly the same pattern emerges by cyclic preservation in *bébe-s* ‘drink.2SG’: during the first cycle, the preaccenting theme vowel of the stem causes stress to fall on the initial syllable, giving rise to a right-aligned bimoraic trochee, which happens to be identical with the default metrical pattern; in the second cycle, however, when the second-person suffix combines with the stem, the preservation of initial stress results in a marked configuration (39a). A more elaborate example is *suplicábais* [su.pli(‘ka.βajs)] ‘drink.PST.IPFV.2PL’ in (29). Similarly, exceptional proparoxytonic stress is generated through faithfulness to an underlying specification in monocyclic *régimen* (10), but emerges by cyclic preservation in polycyclic *cantábamos* [kan(‘ta.βa)mos)] ‘sing.PST.IPFV.1PL’



(39b); see also (30).

|      |                                                     |                                                                                                           |
|------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| (39) | monocyclic                                          | polycyclic                                                                                                |
| a.   | <i>azúcar</i> ‘sugar’                               | <i>bébe-s</i> ‘drink.PRS.2PL’                                                                             |
|      | $[[_{SL} a\theta\acute{u}kar - \{\emptyset, e\} ]]$ | $[[_{SL} [[_{SL} beb - \acute{\sigma}e ] s ]]$                                                            |
|      | $[a(^\circ\theta u.kar)]$                           | $[('be.\beta es)]$                                                                                        |
| b.   | <i>régimen</i> ‘diet, regime’                       | <i>cantá-ba-mo-s</i> ‘sing.PST.IPFV.1PL’                                                                  |
|      | $[[_{SL} réximen ]]$                                | $[[_{SL} [[_{SL} kant - \acute{\sigma}a ] [[_{SL} [b - \acute{\sigma}a ] [[_{SL} mo ] [[_{SL} s ] ] ] ]]$ |
|      | $[('re.xi)men]$                                     | $[kan(^\circ ta.\beta a)mos]$                                                                             |

The principle that all domains affiliated to the same level are subject to the same constraint ranking also entails that the same phonological possibilities will be available to monomorphemic items as to overtly affixed stem-level constructions: Bermúdez-Otero (2024) refers to this as the Monomorpheme Generalization. This too is in agreement with the empirical facts of Spanish stress: the evidence that verified the predictions of Richness of the Base in (36) also corroborates the Monomorpheme Generalization.

In conclusion, our discussion in section 4 has shown that Spanish stress assignment submits to an analysis that upholds a high standard of modularity at the morphosyntax-phonology interface, adopts a completely general theory of cyclic domain assignment, and permits phonological nonuniformity only under strict theoretical control from both Richness of the Base and the prohibition of parallel cophonologies. By adopting such a highly constrained theoretical framework, the analysis predicts (i) that the same set of metrical prespecifications is present in morphs of all types and (ii) that stem-level constructions of all types, whether monocyclic or polycyclic, and overtly affixed or not, are metrically homogeneous. Both predictions are robustly borne out.

## 5 Competing analyses

A number of previous studies, notably including Oltra-Massuet and Arregi 2005, assert that a proper understanding of word-syntax is essential to an adequate description of Spanish stress assignment. We fully endorse this view, but reject the implication that, in consequence, phonology must be granted full access to morphosyntactic structure, with the power to refer to individual syntactic features or heads. We have argued, on the contrary, that the crucial role of word-syntax is channeled by the general theory of cyclic domain assignment. Reference to morphosyntactic information in phonological constraints is restricted to an extremely small set of predicates, including ‘stem’ and ‘heterochromatic’; indeed, our analysis of Spanish uses just the former. In this section, therefore, we focus on two previous studies of Spanish stress in which the relaxation of interface restrictions leads directly to overgeneralization in the form of grammatical statements denoting patterns that are demonstrably not cognitively real.

Following Hooper and Terrell (1976) and Roca (1988), Baković (2016) claims that the default metrical pattern of Spanish is crucially morphologically driven: it places stress on the final syllable of a morphological domain consisting of a stem minus its final thematic element. This proposal incurs a range of problems. We pass over the difficulties arising from the fact that Baković's notion of “stem” rests on an incorrect analysis of the underlying distribution of theme vowels: (40a), instead of correct (40b); see Bermúdez-Otero 2006, 2013:21-57 for extensive argumentation. These difficulties may at any rate be obviated by restating Baković's generalization as in Oltra-Massuet and Arregi 2005:64 (40c).

- (40)        *man-áz-a* [ma. 'na.θa] ‘hand.AUG’
- a.         $\llbracket \llbracket \text{man}_\sqrt{\phantom{x}} - \emptyset_n \rrbracket \text{a}\theta_{\text{AUG}} \rrbracket \text{a}_{\text{TH}}$
- b.         $\llbracket \llbracket \text{man}_\sqrt{\phantom{x}} - \emptyset_n - \text{o}_{\text{TH}} \rrbracket \text{a}\theta_{\text{AUG}} - \text{a}_{\text{TH}} \rrbracket$
- c.        Stress falls on the vowel preceding TH.

The main problem arises over the fact that Baković's analysis requires a large set of word-final unstressed vowels to be analysed as thematic elements in the absence of any independent justification. This includes the final vowels of the citation forms of function words, such as degree modifiers (e.g., *cási* (3b)), prepositions (e.g., *pára* (3c)), and wh-items (e.g., *cuándo* (3c)); in none of those items does the final vowel participate in the exponence of any syntactic feature, unlike the theme vowels of nouns, adjectives, and verbs. Similarly, Baković's analysis requires that we parse as thematic elements the final vowels of acronyms (4a) and truncata (41); yet in the latter we observe that truncation proceeds in exactly the same manner whether or not the final vowel of the truncatum belongs in the set of vowels that perform exponence roles in nominal morphology (i.e., *-o*, *-a*, and *-e*).

|      |                        |                      |             |
|------|------------------------|----------------------|-------------|
| (41) | <i>móto</i> ['mo.to]   | < <i>motocicleta</i> | ‘motorbike’ |
|      | <i>cómpa</i> ['kom.pa] | < <i>compañero</i>   | ‘buddy’     |
|      | <i>cóle</i> ['ko.le]   | < <i>colegio</i>     | ‘school’    |
|      | <i>póli</i> ['po.li]   | < <i>policía</i>     | ‘police’    |
|      | <i>cómpu</i> ['kom.pu] | < <i>computadora</i> | ‘computer’  |

Conversely, words such as *canapé* ‘canapé’ (7b) and *tabú* [ta.‘βu] ‘taboo’ are to be parsed (correctly) as ending in vowels that belong to the root. Thus, Baković's analysis becomes viciously circular: a stem-final vowel is thematic if unstressed, and unstressed if thematic.

In consequence, Baković's generalization goes badly astray if the application of the label ‘thematic element’ is constrained by some independent criterion. For example, if we parse as theme vowels only those that are observed to participate in the exponence of some syntactic feature in some context, then (40c) vastly overgeneralizes the marked pattern of oxytonic stress on a light

ultima: it predicts \**casí*, \**UCÍ*, \**compú*, etc. Alternatively, one might adopt an entirely grammatically vacuous notion of theme vowel, according to which all stem-final unstressed vowels are thematic. In that case, however, Baković's formulation of the default metrical pattern of Spanish converges empirically with ours, but with the unjustified additional cost of abandoning modularity. This additional cost becomes manifest in the fact that our formulation of the default pattern (i.e., align a bimoraic trochee with the right edge of the domain) is typologically natural (Hayes 1995:181-182), whereas Baković's is entirely parochial: it arbitrarily positions stress by reference to a morphological object, the thematic element, which is absent in many languages.

Oltra-Massuet and Arregi's (2005) account of stress in nonverbal items suffers from exactly the same drawbacks as Baković's, since it adopts rule (40c). In addition, their analysis of verb stress incurs a different set of problems. In this case, we pass over the difficulties arising from their assumption that stress is computed in a single cycle over the entire inflected wordform (see also Myler 2015; cf. Bermúdez-Otero 2016:408-413). This fails to account for the facts of high vocoid syllabification illustrated in (15, 16), but we shall assume that an alternative version of Oltra-Massuet and Arregi's proposals exists in which verb stems do trigger cycles. The main problem arises over the fact that Oltra-Massuet and Arregi formulate the default metrical pattern for verbs as placing stress immediately to the left of the T head. The acid test for this generalization is found in the present tenses, where, as we saw in section 3.3, the columnar behavior prevalent elsewhere fails (31). In the absence of a columnar pattern, some present-tense forms must be identified as exhibiting the default pattern, the others as displaying a marked pattern driven by a more specific mechanism. Here, Oltra-Massuet and Arregi's analysis clearly identifies the 2SG, 3SG, and 3PL forms as marked: in (42), they are highlighted with a manicule as being cases in which stress unexpectedly falls further to the left than the vowel immediately preceding the fused T/AGR head. In contrast, 1SG, 1PL, and 2PL forms do show Oltra-Massuet and Arregi's default.

(42) ‘sing.PRS.IND’ (see Oltra-Massuet and Arregi 2005:59, table 7)

|     | √     | ν | T <sub>H</sub> | T/AGR |   |
|-----|-------|---|----------------|-------|---|
| 1SG | 'kant | ∅ | ∅              | o     |   |
| 2SG | 'kant | ∅ | a              | s     | ☞ |
| 3SG | 'kant | ∅ | a              | ∅     | ☞ |
| 1PL | kant  | ∅ | 'a             | mos   |   |
| 2PL | kant  | ∅ | 'a             | js    |   |
| 3PL | 'kant | ∅ | a              | n     | ☞ |

In contrast, the analysis set out in this article makes diametrically opposed predictions. According to our proposals, inflected verbs bear stress on the rightmost accented vowel in the input to the stem-level cycle defined by the whole form. In the present tense, this is expected to be the last vowel of the root, as the theme vowel adjoined to  $\nu$  is preaccenting and the exponents of the fused T/AGR head are too small to project foot-heads by themselves in their respective cycles. Consequently, stress on the last vowel of the root emerges as the default pattern for present-tense verb forms. We derive the unexpected stress profile of 1PL and 2PL forms by stipulating that the 1PL and 2PL subject agreement markers have preaccenting allomorphs contextually restricted to the present tenses (32a). In (43), the manicule highlights the marked forms containing these special allomorphs.

(43) ‘sing.PRS.IND’ (see (31))

|     | √    | v | TH  | T/AGR  |   |
|-----|------|---|-----|--------|---|
| 1SG | kant | ∅ | σ̣a | o      |   |
| 2SG | kant | ∅ | σ̣a | s      |   |
| 3SG | kant | ∅ | σ̣a | ∅      |   |
| 1PL | kant | ∅ | σ̣a | σ̣mo-s | → |
| 2PL | kant | ∅ | σ̣a | σ̣i-s  | → |
| 3PL | kant | ∅ | σ̣a | n      |   |

Fortunately, a few dialects of the Spanish language have performed a natural experiment in which the default stress pattern of the present tense is generalized throughout the present subjunctive paradigm, eliminating exceptions and enforcing a columnar pattern. The fact that this natural experiment is restricted to the present subjunctive, and does not take place in the indicative, is straightforwardly explained by frequency: the subjunctive mood is rarer in discourse than the indicative, and so provides the learner with fewer cues to exceptional behavior (e.g., Pinker 1999, Bybee 2001). In (44) we lay out the present subjunctive paradigm of Uruguayan Spanish (Martínez-Paricio 2013:135). Note that this is one of the varieties where 2PL is syncretic with 3PL. In (44) the manicule highlights the 1PL form, which has undergone historical change, diverging from the realization found in more conservative dialects.

- (44) 'love.PRS.SBJV'
- |     |             |                                            |
|-----|-------------|--------------------------------------------|
| 1SG | ['a.me]     |                                            |
| 2SG | ['a.mes]    |                                            |
| 3SG | ['a.me]     |                                            |
| 1PL | ['a.me.mos] | ☞ cf. [a.'me.mos] in conservative dialects |
| 3PL | ['a.men]    |                                            |

Clearly, historical change has taken place in a form which Oltra-Massuet and Arregi designate as exhibiting the default pattern, and which our analysis classes as exceptional. Since paradigm levelling extends default patterns at the expense of exceptions and not vice versa, the evidence of (44) refutes Oltra-Massuet and Arregi's analysis of present-tense stress and confirms ours.

In fact, Oltra-Massuet and Arregi are aware of these data (2005:60, fn. 22). They describe them as involving a reanalysis of the thematic /-e-/ of the subjunctive, which allegedly goes from being adjoined to *v* to being adjoined to T. No learning-theoretic account is proposed to account for this scenario, and the role of frequency is not addressed. In our analysis, in contrast, the change can be easily understood as driven by the learner's assumption that the less specific allomorph applies, except in contexts providing sufficient positive evidence for a more specific option. In the dialects under discussion, probably partly owing to the syncretism of 2PL with 3PL, the evidence for the presence of preaccenting allomorphs of AGR markers in the present tense continues to reach the requisite threshold for acquisition in the high-frequency indicative mood, but has fallen below it in the low-frequency subjunctive.

In sum, both Baković (2016) and Oltra-Massuet and Arregi (2005) propose default patterns of stress assignment directly referring to specific syntactic positions, thus vastly relaxing modularity restrictions at the morphosyntax-phonology interface. The increased power of their grammatical systems thus enables them to make generalizations that would otherwise remain unavailable. The

evidence presented in this section shows that this power increase is not justified: those generalizations are not cognitively real. On the contrary, our analysis demonstrates that a fully adequate treatment of Spanish stress is available within a framework that upholds a high standard of modularity, provided that the crucial role of word-syntax is channeled through an adequate procedure for cyclic domain assignment.

## 6 Conclusions

Spanish stress assignment has prompted analyses of formidable intricacy (e.g., Roca 2020). In the light of this article, however, the overall pattern appears simple. Initially, minimal stems and affixes are metrified individually. At this stage, a bimoraic trochee is built at the right edge of each domain unless prevented by an underlying prespecification or, in the case of affixes, by subminimal size. High-ranking faithfulness constraints at the stem level protect only a small set of crucial underlying prespecifications, confined to the final three-syllable window. In subsequent stem-level cycles, stress falls on the rightmost accented syllable in the input within the final trisyllabic window. At the word-level, the same system remains in operation, except that the right-edge window is switched off. That is all.

This straightforward analysis is enabled by a highly restrictive theoretical framework. Phonological constraints do not refer to morphosyntactic properties at all, except for an extremely small set of predicates such as ‘stem’ and ‘heterochromatic’, of which the analysis uses only the former. Morphosyntactic conditioning is thus largely channeled indirectly through a hierarchy of cyclic phonological domains; these are defined by a general (and indeed plausibly universal) procedure, once the idiosyncratic affiliation of individual affixes to the stem or word level is known. Nonuniformity is limited to (i) the lexical prespecifications protected by high-ranking faithfulness constraints and (ii) the binary choice between the stem-level and word-level phonology. Each stratum is restricted to a single constraint ranking. This is enough; no more power is required.



Because it is couched in a highly restrictive theoretical framework, the analysis generates rich empirical predictions, which we have been able to verify. First, we have seen that the same set of metrical prespecifications is available to stems and suffixes of all types; in particular, we have shown that theme vowels come in three different metrical flavors, as unaccented, accented and preaccenting. Secondly, we have shown that the same range of marked metrical patterns that occur in minimal stems through lexical prespecification also arise in polycyclic stem-level items through cyclic preservation (Chung's Generalization). An analogous uniformity holds between monomorphemic and overtly affixed stem-level constructions (Monomorpheme Generalization). Alternative analyses that adopt more relaxed approaches to modularity and nonuniformity fail to recover these predictions and permit the formulation of generalizations which, upon inspection, prove not to be cognitively real. This result strongly supports the specific package of foundational assumptions adopted in this article, comprising qualified principles of modularity, cyclicity, and constrained nonuniformity.

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